

Ada Boost Machine Learning Algorithms

Decision Trees

{ Base learners }

↓
Bagging

- ① Random Forest Classifier
- ② Random Forest Regressor

Ensemble Techniques

{ weak learners }

↓
Boosting

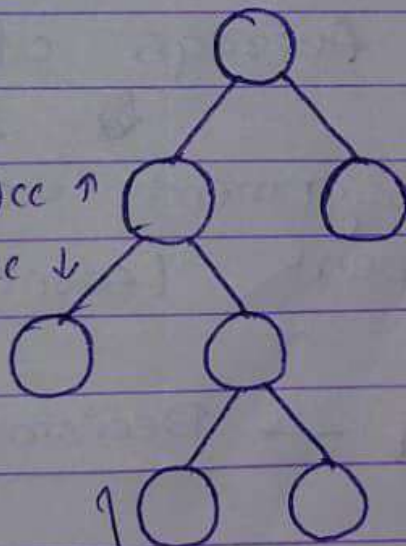
- ① Adaboost
- ② Gradient Boosting
- ③ Xgboost

Decision Tree :

Overfitting :-

Train Acc ↑

Test Acc ↓



Low Bias

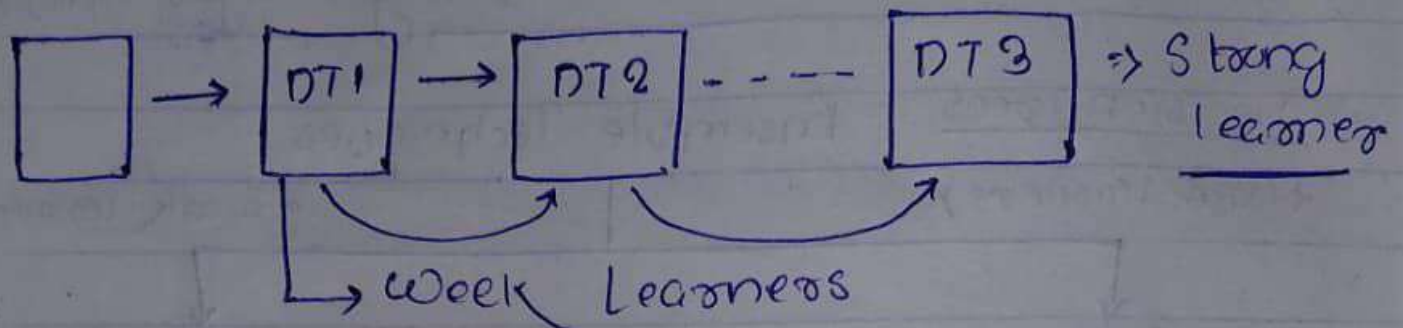
High Variance

Random Forest

{ Bagging }

{ Low Bias
Low Variance }

Boosting & sequentially connected



Weak learner → Haven't learn much from the Training Dataset.

Random Forest → Majority voting Classifier
Average of (O/P)

Ada Boost → Assignment weights to the weak learners

$M_1, \dots, M_n \rightarrow$ Decision Tree Stumps

$$f = \alpha_1(M_1) + \alpha_2(M_2) + \alpha_3(M_3) + \dots + \alpha_n(M_n)$$

$\{\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n\} \Rightarrow$ weights

→ CLASSIFICATION
→ REGRESSION

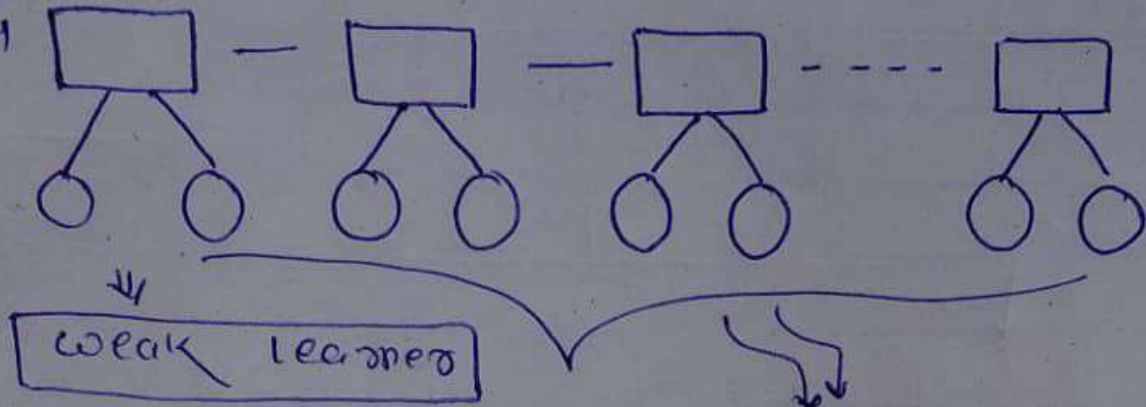
Decision

Stump

means depth is 1

87

Depth = 1



↳ Underfitting

↳ Train Acc ↓ 40%

↳ Test Acc ↑ 45%

Decision Tree Stump

↓
High Bias / Low Variance → Low Bias / High Variance